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SUPERSEDING
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MILITARY SPECIFICATION

HAMMERS, TRACK SPIKE, GASOLINE- OR DIESEL-ENGINE-DRIVEN

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers power-driven spike hammers for driving spikes into railroad ties.

1.2 Classification. Hammers shall be of the following types and classes as specified (see 6.2):

Type I - 36-inch (914 mm) Gage.
Type II - 39-3/8-inch (1000 mm) Gage.
Type III - 42-inch (1067 mm) Gage.
Type IV - 56-1/2-inch (1435 mm) Gage.
Type V - 60-inch (1524 mm) Gage.
Class 1 - Gasoline-Engine-Driven.
Class 2 - Diesel-Engine-Driven.

FSC 2230

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

O-E-915	- Extinguisher, Fire, Dry-Chemical (Hand-Portable).
QQ-S-781	- Strapping, Steel, and Seals.
UU-T-81	- Tags, Shipping and Stock.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-P-3321	- Painting and Marking: Railway Motive Power and Work Equipment.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-T-22085	- Tapes, Adhesive, Preservation and Sealing.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-B-26195	- Boxes, Wood-Cleated, Skidded, Load-Bearing Base.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-L-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-F-52723	- Filter Element, Hydraulic, Disposable.

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STANDARDS

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1474	- Noise Limits for Army Materiel.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Standard and Recommended Practices.

J512 - Automotive Tube Fittings.
J514d - Hydraulic Tube Fittings.
J516 - Hydraulic Hose Fittings.
J517a - Hydraulic Hose.
J531 - Automotive Pipe, Filler and Drain Plugs.
J532 - Automotive Straight Thread Filler and Drain Plug.
J534 - Lubricating Fittings.
J551 - Measurement of Electromagnetic Radiation from a Motor Vehicle or Other Internal-Combustion Powered Device (Excluding Aircraft) (20-1000 MHz).
J744 - Hydraulic Power Pumps.
J925 - Minimum Access Dimensions for Construction and Industrial Machinery.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096).

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NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking ssociations, Inc., ATTN: Traffic Department, 1616 P Street NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

AMERICAN WELDING SOCIETY, INC.

Standard Qualification Procedure.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Environmental Protection Agency (EPA) Control of Air Pollution From New Motor Vehicles and New Motor Vehicle Engines.

(Application for copies should be reference Code of Federal Regulations, 45 CFR, Part 1201 and should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. The track spike hammers shall be portable, gasoline- or diesel-engine-driven, hydraulic-powered units for driving spikes into track ties. The units shall be mounted on four ductile iron or cast steel single-flanged wheels for transit on track. The frame shall be constructed

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of steel and shall include a transversing rail for support and positioning of the spike driving assembly. The hammers shall require only one man for operation and shall be capable of driving spikes on both sides of the rails while working in the same direction. The hammers shall drive spikes at a rate of at least 5 spikes per minute. Spikes shall be driven into the ties close to the base of the rail, securing the rail to the ties without bending or twisting the spikes.

3.1.1 Drawings. The contractor is responsible for preparing his own engineering drawings. No deviation from the prescribed dimensions is permissible without prior approval of the contracting officer.

3.2 First article (first-produced hammer). The contractor shall furnish a hammer within the time frame specified in the contract, to prove prior to starting production, that his production methods and choice of design detail will produce a hammer that complies with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government.

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.10).

3.3.1 Screw threads. All screw threads shall be in accordance with Handbook H28.

3.3.2 Welding. Unless otherwise specified, all welding shall be in accordance with the latest applicable codes of the American Welding Society.

3.4 Interchangeability. Parts and components shall be interchangeable between hammers.

3.5 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, or replacement without the removal of other major assemblies and installed attachments. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners. Drain outlets shall be located for accessibility. Provisions shall be made to permit the use of receptacles for collecting drainage.

3.6 Safety. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions.

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3.6.1 Human factors. The characteristics of the hammer shall provide for operation by personnel ranging from small man (5th Percentile) clothed, through the large man (95th Percentile), arctic-clothed, in accordance with SAE J925. The noise level at the operator's station shall not exceed the limits specified in MIL-STD-1474, Category C, for steady-state noise when tested in accordance with 4.5.2.7 (aural nondetectability and impulse noise do not apply). The precautions of MIL-STD-1474 shall be provided if the sound level exceeds 85 dB(A) when tested in accordance with 4.5.2.7. Hearing protectors shall not be supplied with the hammer. Hazard sign, when required, shall state "CAUTION, HEARING PROTECTION REQUIRED".

3.7 Overall dimensions and weight. The overall dimensions and weight of the hammer in travel position shall be as follows:

Weight with setoff	2080 pounds (945 kg) (Nominal)
Length	72 inches (1 830 mm) (Nominal)
Width	96 inches (2 440 mm) (Nominal)
Height	54 inches (1 370 mm) (Nominal)

3.8 Bearings. All bearings shall be of the antifriction type that will carry the radial and axial loads.

3.9 Turntable. The hammer shall be furnished with a built-in center lift turntable. The turntable shall rotate freely 360 degrees.

3.10 Setoff. A folding-type setoff shall be furnished and stored on the hammer. The setoff shall weigh not more than 100 pounds (45 kg). Not more than two men shall be required to remove the hammer from the rails and place the hammer to the side and clear of the track.

3.11 Braking system. The hammer shall be equipped with a hydraulic or mechanical braking system that will hold the hammer in a stationary position when brakes are applied.

3.12 Hydraulic system. The hydraulic system shall provide a means of operating the spike hammer. Oil for the system shall conform to MIL-L-2104, Grade 10. For low temperature operation, oil conforming to MIL-L-46167 may be used. All hydraulic component materials shall be compatible with these oils. Unless otherwise specified herein, all system components shall withstand a proof pressure that is equal or greater than 1-1/2 times the maximum working pressure (1-1/2 times relief valve setting) without external leakage, damage, or permanent deformation. Hydraulic connection ports of the hydraulic system components shall be sealed by the component manufacturer and this seal shall be maintained until final assembly on the hammer.

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3.12.1 Test points. One test point shall be located at the outlet of each pump for measuring pressure.

3.12.2 Cylinders. The cylinders shall be fabricated from seamless steel tubing, or welded and redrawn tubing. Each cylinder shall have an exclusion device at the rod to prevent foreign material and fluids from entering and damaging the seal and other cylinder components. The cylinders shall be located as close as practicable to the track wheels. The piston rod shall be hard chrome plated to prevent corrosion. The bearing surface shall be a material that will not scar, score, or gall the inside surface of the cylinder.

3.12.3 Pump. The hammer shall be equipped with hydraulic pump(s) driven directly by a gasoline- or diesel-engine. The pump(s) shall be a constant displacement or variable displacement type and shall be capable of developing the system relief valve setting pressure at a speed corresponding to the manufacturer's rated speed. The pump(s) shall be in accordance with SAE J744.

3.12.4 Controls. All hydraulic control valves shall be of the metering type so that accurate control of the unit may be maintained at all times. Pushing of a lever shall result in a lowering function and pulling on shall result in a raising or retracting function. The travel of the controls shall not exceed 6 inches (152 mm). The travel control grip shall have different configuration from that of the spike driver control levers. All ıtrol-valve spools shall be spring centered with sufficient force to ˌcurn the levers to the center position and retain it there throughout all operations. Valve spools exposed to dirt and moisture shall be provided with wipers and seals.

3.12.5 Relief valve. A relief valve shall be provided to limit the pump pressure to the maximum required to perform the necessary system functions. Pressure equal to the relief valve setting shall be obtained only when a cylinder is bottomed or when the system is subjected to shock loading.

3.12.6 Return line filter. The system filler element(s) shall conform to MIL-F-52723. The filter shall maintain a full flow condition during all operations of the hammer. The filter bypass may open a system oil at a temperature less than 70° F (21° C). All return oil shall pass through the filter except relief oil which may return directly to the reservoir. An automatic bypass that permits full flow at a pressure of not less than 35 psid (241 kPa) shall be provided in the filter housing. The filter shall be equipped with a contamination indicator which may be remotely mounted. The indicator shall show when the element is clean and, as a minimum, when the filter is beginning to bypass, requiring a change of the element.

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3.12.7 Reservoir. The reservoir shall include a filler opening, screen, cap, and other items required to function as specified herein. The reservoir shall be rigidly mounted to prevent vibrations that may result in leaks, loosening of fasteners, or cracks. Connections on the reservoir shall be located so that connecting lines can be removed without removing the reservoir or any other major component. The reservoir should be baffled when necessary to provide a tortuous flow path. A dipstick or sight gage shall be furnished to determine the reservoir oil level. When practical, the sight gage or window should be visible to the operator when he is in his normal position. The reservoir pump suction openings should be located a minimum of 1 inch (25 mm) above the bottom of the reservoir to prevent entrance of settled contaminants. Except for sealed reservoir, a breather shall be provided and shall filter incoming air.

3.12.8 Hydraulic lines. Hydraulic lines shall consist of tubing or pipe, hose, and fittings. Hose or a combination of hose and tubing shall be used between components that move in relation to each other. All lines shall be routed to provide the shortest line, minimum number of bends, ease of maintenance, and maximum protection. The hose shall be supported, when necessary, to eliminate sagging at the fitting.

. 3.12.8.1 Pressure hose assemblies. Pressure hose shall be as specified in SAE J517 and pressure hose couplings shall be as specified in SAE J516. The hose selected shall have a working pressure which is equal to or greater than the hydraulic system maximum working pressure (relief valve setting).

3.12.8.2 Pressure tubing and piping. Pressure tubing and piping shall have a minimum burst pressure rating of 3.5 times the system maximum working pressure relief valve setting. All tubing bends shall be smooth without flattening, kinking, or wrinkling of the tube. All tubing and piping shall be cleaned free of foreign matter including cuttings and weld slag and spatter.

3.12.8.3 Pressure pipe fitting and pressure tube fittings. Unless otherwise specified herein, pressure tube fitting shall conform to either SAE J512 or SAE J514. Pressure tube fitting shall have burst pressure rating that is equal to or greater than 3.5 times the maximum system working pressure (relief valve setting).

3.13 Propelling unit. The hammer shall be self-propelled or hydraulically operated and shall move in forward and reverse directions by means of a 2-speed transmission.

3.14 Spike-driver assembly. The hammer shall be equipped with a spike-driver assembly connected to the hydraulic pump with a flexible hose. The driver shall be mounted in a carriage that rides the length of the trans-

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verse track on metal rollers. The driver shall have a 5-inch (127 mm) minimum longitudinal movement for positioning over the spike to be driven when the hammer is stationary.

3.15 Engine. For the purpose of selecting the engine, the hammer shall be classified as a Class I end item in accordance with MIL-STD-1410. The engine shall be gasoline or diesel as specified and shall conform to MIL-STD-1410, Class I unless otherwise specified herein. All systems, components, and accessories shall conform to MIL-STD-1410 unless otherwise specified herein.

3.15.1 Fuel tank. The capacity of the tank shall be sufficient for 8 hours of continuous operation in operating condition specified herein.

3.15.2 Accessories. The engine shall be furnished with the following accessories: A housing, an electrical cranking system and an instrument panel consisting of but not limited to the following instruments, as applicable:

(a) Gage, lubricating oil.
(b) Indicator, fuel, liquid capacity.
(c) Tachometer.
(d) Hourmeter.
(e) Temperature gage.

3.15.3 Clutch. The clutch shall be located between the engine and transmission. The clutch shall be rated by its manufacturer at not less than 150 percent of the maximum torque developed by the engine.

3.15.4 Exhaust system. The engine shall be equipped with a muffler and an exhaust pipe located and installed in such a manner that exhaust gases will not be discharged under the hammer or be hazardous to operating personnel.

3.15.5 Crankcase ventilation system. A positive crankcase ventilation system in accordance with EPA requirements shall be provided on each engine. If the ventilation system is connected to the air cleaner, it shall be connected in such a manner that no vapors or condensation will come in contact with, or be deposited on the air cleaner element.

3.16 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.16.1 Lubricants. The hammer, including hydraulic system, shall operate as specified herein when lubricated with military lubricants, or as recommended by the manufacturer for this type of equipment (see 6.6).

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3.16.2 Lubrication fittings. Lubrication fittings, when used, shall conform to SAE J534. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun and a flexible extension.

3.16.3 Enclosures with integral reservoir of lubricants. Enclosures such as gear cases and transmission housings which contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with a lubricant and for complete draining. The drain outlet shall be fitted with a magnetic drainplug conforming to SAE J531 and J532 size 7/8-14UNF-3.

3.16.4 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as recommended by the manufacturer or as follows:

(a)	MIL-G-23827	- For instruments, all sealed bearings, and other bearings where operating temperatures do not exceed 250° F (121° C).
(b)	MIL-G-10924	- For bearings where operating temperatures do not exceed 175° F (79° C).

Each hammer shall be assembled, run-in (broken-in) for 10 minutes and delivered with these greases. Each component shall be tagged in a conspicuous place to indicate which grease has been used.

3.16.5 Oil lubrication. Unless otherwise preserved for packaging (see 5.2), all components shall be filled to the operating level with winter grades of military or commercial oils designated for use in the temperature range of 0° F to 32° F (-18° to 0° C). Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used.

3.17 Fire extinguisher. A 2-1/2-pound (1 kg) fire extinguisher conforming to O-E-915, Type III, Class 2, with mounting bracket, shall be furnished and installed on each hammer. The extinguisher shall be mounted in a position accessible to the operator.

3.18 Lifting attachments. Permanently affixed attachments that enable the hammer to be lifted in its normal travel position shall be provided. Each attachment shall withstand, without damage to any part of the hammer, 2-1/2 times the force imposed when the hammer is suspended by the lifting attachments. Attachments for multiple slings shall be located so that:

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(a) The slings will converge not more than 19 feet (6 m) above the lowest extremity of the hammer and
(b) Not less than 1-inch (25 mm) clearance will be maintained between the slings and the hammer.

The lifting post head shall be furnished with a cored hole for attachments. Lifting attachments may also be used as tiedown attachments (see 3.19).

3.19 Tiedown attachments. Permanently affixed attachments that permit fastening the hammer to the floor or deck of a transportation medium shall be provided. Each attachment shall withstand, without damage to the hammer, its proportion of the following static loads: 8-G forward, 8-G rearward, 2-1/2-G upward, 1-1/2-G sideward, where G equals cargo weight of the hammer. Attachments shall be rounded and beveled to prevent cutting of the tiedown material. Tiedown attachments may also be used as lifting attachments and hand rails may be used as tiedown attachments (see 3.18).

3.20 Tie nipper attachment. When specified (see 6.2), a tie nipper attachment for the hammer shall be furnished. The nipper shall connect to and operate from the hydraulic system and have sufficient power to raise and hold the tie firmly against the rail for spiking operations. The nipper attachment shall not increase the weight of the hammer more than 300 pounds (135 kg). The pedestals on the nipper reaction arms shall lock in the up position when deadheading.

3.21 Identification marking. The hammer shall be identified in accordance with MIL-STD-130. Identification plates shall conform to MIL-P-514, Type I, Style 2, photographic plate. Transportation data plates shall conform to MIL-P-514, Type I, Style 4, photographic plate, and shall indicate location and capacity of lifting and tiedown attachments.

3.22 Instruction plates. Each hammer shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedure to be followed in operating or servicing the hammer. Instruction plates shall conform to MIL-P-514, Type I, Style 3, photographic plate.

3.23 Treatment and painting. The portions of the hammer and its components and parts normally painted shall be cleaned, treated, and painted in accordance with MIL-P-3321.

3.24 Toolbox. Each hammer shall be provided with a weatherproof toolbox The lid shall remain in the fully open position either by its own weight or by holding devices. A hasp which will accommodate a key-operated lock shall be provided for locking the toolbox. The toolbox shall be of a size to hold the maintenance tools and repair parts specified.

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3.25 Workmanship.

3.25.1 Steel and aluminum fabrication. Steel and aluminum used in the fabrication of the hammer shall be free from kinks and sharp bends. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape.

3.25.2 Bolted connections. Boltholes shall be punched or drilled and shall have the burrs removed. Washers shall be provided in accordance with standard railway practice.

3.25.3 Riveted connections. Rivet holes shall be punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet holes shall be full, neatly made, concentric with other rivet holes, and in full contact with the surface of the adjacent member.

3.25.4 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.

3.25.5 Machine work. Tolerance and gages for metal fits shall conform to the limitations specified herein and otherwise within manufacturing tolerances to permit correct fittings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First-produced hammer inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

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4.3 First-produced hammer inspection.

4.3.1 Examination. The first-produced hammer shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first-produced hammer shall be tested as specified in 4.5.2 through 4.5.2.7. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each hammer shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each hammer shall be tested as specified in 4.5.2.3. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The hammer shall be examined for the following defects:

101. Wheels for the railway hammer not as specified.
102. Material not as specified.
103. Screw threads not as specified.
104. Welding not as specified.
105. Safety not as specified.
106. Human factors not as specified.
107. Overall dimensions and weight not as specified.
108. Bearings not as specified.
109. Braking system not as specified.
110. Tie nipper not as specified.
111. Hydraulic system not as specified.
112. Test points not as specified.
113. Cylinders not as specified.
114. Pumps not as specified.
115. Controls not as specified.
116. Relief valve not as specified.
117. Return line filter not as specified.
118. Reservoir not as specified.
119. Hydraulic lines not as specified.
120. Pressure hose assemblies not as specified.
121. Pressure tubing and piping not as specified.
122. Pressure pipe fitting and pressure tube fittings not as specified.
123. Propelling unit not as specified.

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124. Engine not as specified.
125. Fuel tank not as specified.
126. Accessories not as specified.
127. Clutch not as specified.
128. Exhaust system not as specified.
129. Crankcase ventilation system not as specified.
130. Operator's seat not as specified.
131. Lubrication not as specified.
132. Lubricants not as specified.
133. Lubrication fittings not as specified.
134. Pressure-release device not as specified.
135. Enclosures with integral reservoir of lubricants not as specified.
136. Grease lubrication not as specified.
137. Oil lubrication not as specified.
138. Fire extinguisher not as specified.
139. Lifting attachments not as specified.
140. Tiedown attachments not as specified.
141. Identification marking not as specified.
142. Instruction plates not as specified.
143. Treatment and painting not as specified. Toolbox not as specified.
144. Steel and aluminum fabrication not as specified.
145. Bolted connections not as specified.
146. Riveted connections not as specified.
147. Welding not as specified.
148. Machine work not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Prior to test, the hammer shall be lubricated with oils and greases specified in 3.16 and run-in for 10 minutes exercising all functions without load. Oils shall be those designated for use in the ambient temperature at the place of test. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The engine shall be governed at a speed not greater than the continuous rated speed as established by the qualification test. The hydraulic pumps shall not exceed the continuous rpm recommended by the pump manufacturer when operating at not more than 2000 psi (13 800 kPa).

4.5.2.2 Hydraulic system. The hydraulic components, except reservoir, shall be pressurized to not less than 1-1/2 times the maximum working pressure (1-1/2 times relief valve setting) required to meet the requirements

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of this specification. The reservoir shall be pressurized to 3.5 psi (0.11865 (Kg/m^2). Any external leaks, deformation, or breakage of any component shall constitute failure of this test.

4.5.2.3 Operational test. The hammer shall be operated for not less than 15 minutes performing all operational functions as specified herein under specified load. Any operational difficulties, breakage, or evidence of contamination on hydraulic filter, leakage, or permanent deformation shall constitute failure of this test.

4.5.2.4 Spike driving. The spike driving test shall be executed by one man. Drive spikes into ties at a rate of not less than five spikes per minute for a distance of 1/2 mile (0.8 km). The spikes may be prestarted in the ties and four spikes shall be driven in each tie, one on each side of both rails. Inability to drive spikes at a rate of five per minute into the ties on the specified gage track; inability to drive the spikes close to the rail without bending or twisting of the spikes, or breaking or chipping of the rail; or failure of the brakes to hold the hammer stationary while being operated shall constitute failure of this test.

4.5.2.4.1 Spike driving with tie nipper attached. When a tie nipper attachment is specified (see 3.20), using the hammer with tie nipper attached, nip, raise, and hold 12 ties firmly against the rail during the spike driving test specified in 4.5.2.4. Inability to nip and hold the 12 ties f ly against the rail while driving spikes shall constitute failure of t test.

4.5.2.5 Setoff. Remove the hammer from the railroad track onto the setoff rails and move to the side and clear of the track using only two men and the built-in turntable. Then reverse the operation and set the hammer back on the railroad track. Inability to complete the test in not more than 5 minutes shall constitute failure of this test.

4.5.2.6 Lifting and tiedown attachments. The vectorial force as specified in 3.18 for lifting attachments and the static loads as specified in 3.19 for tiedown attachments shall be applied to the attachment when mounted on its structural member. Weld failure or permanent deformation of the attachment, the fastening, or the structural member shall constitute failure of this test.

4.5.2.7 Noise level measurements. Measure the noise at the operator's station in accordance with MIL-STD-1474 for steady-state noise while operating the jack. Nonconformance to 3.6.1 shall constitute failure of the test.

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4.6 Inspection of packaging.

4.6.1 Preproduction pack inspection.

4.6.1.1 Examination. Examine the preproduction pack for the defects specified in 4.6.2.2. Presence of one or more defects shall be cause for rejection.

4.6.1.2 Test. The preproduction pack of a boxed hammer for Level A or Level B shall be subjected to the railroad car test as specified for guided impact tests in accordance with MIL-STD-1186, Appendix A. The car shall strike a string of five empty cars with draft gear extended and the brakes set, at a speed of not less than 10 miles (16 km) or more than 11 miles (18 km) per hour. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking or bracing, or any visual damage to the components shall constitute failure of the preproduction pack and shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Examination. Each unit of product shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

149. Materials, methods, and containers not as specified for Level A or B. Each incorrect material, method, or container shall be considered one defect.
150. Disassembly not as specified for Level A or B.
151. Unpainted exterior metal surfaces not coated with preservative for Level A or B.
152. Engine and engine accessories not preserved in accordance with the referenced document for Level A or B.
153. Liquid cooling systems not preserved as specified for Level A, B, or C.
154. Hydraulic system not preserved as specified for Level A or B.
155. Gears not preserved as specified for Level A or B.
156. Drive chain not preserved as specified for Level A or B.
157. Waterproof ink not used on information tags for Level A.
158. Instrument panel not protected as specified for Level A or B.
159. Repair parts not preserved in accordance with the referenced document for Level A or B.
160. Technical publications not preserved as specified for Level A or B.

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5.2.1.4 Engine. The engine and engine accessories shall be preserved in accordance with MIL-E-10062, Type II, Level A, Method I. If a liquid-cooled engine is furnished, the cooling system shall be preserved by the water and antifreeze procedure, except the antifreeze shall conform to MIL-A-46153.

5.2.1.5 Hydraulic system. The hydraulic fluid supply reservoir shall be filled with the hydraulic fluid recommended for operation. The pistons shall be retracted as far as practicable into the cylinders and secured. Any exposed portions of the piston rods shall be coated with Type P-6 or P-11 preservative and wrapped or covered with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2. The wrap shall be secured in place with tape conforming to MIL-T-22085, Type I or II. When the pistons can be fully retracted, any remaining uncoated surfaces of the piston rods shall be coated with Type P-1 preservative, with no wrapping required. The hydraulic control valves shall be secured in the neutral position and preserved as specified for the piston rods. Hydraulic lines shall not be disconnected. A tag conforming to UU-T-81, Type A, shall be attached in a conspicuous location indicating: "The hydraulic system is filled with the fluid recommended for operation." The information shall be applied on the tag with waterproof ink.

5.2.1.6 Gears.

5.2.1.6.1 Exposed gears. Unpainted surfaces of exposed gears shall be coated with Type P-1 preservative.

5.2.1.6.2 Enclosed gears. Gear housings requiring a SAE 10, 30, or 50 lubricating oil shall be filled to the operating level with preservative lubricating oil conforming to P-10, Type I, Grade 10, 30 or 50 as applicable. A tag shall be attached in a conspicuous location indicating: "The housing contains preservative oil good for operation. Do not drain until the first recommended lubricant change." Gear housings requiring a lubricating oil other than SAE 10, 30 or 50 shall have the housing filled to the operating level with lubricant recommended for operation. A tag shall be attached indicating: "The housing is filled with the operational lubricant. Do not drain until the first recommended lubricant change." Tags shall conform to UU-T-81, Type A, and the information shall be applied with waterproof ink.

5.2.1.7 Drive chain. Exposed surfaces of drive chain shall be coated with enough Type P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After the excess preservation has drained, the chain shall be coated with Type P-1 preservative. Sprockets shall be coated with Type P-1 preservative.

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161. Maintenance tools not preserved in accordance with the referenced document for Level A or B.
162. Consolidation not as specified for Level A or B.
163. Packing not as specified for Level A or B.
164. Preservation and depreservation guide not prepared as specified for Level A, B, or C.
165. Marking missing, illegible, incorrect, or incomplete for Level A, B, or C.

5. PACKAGING

5.1 Preproduction pack. The contractor shall furnish a preproduction pack for examination and test of the boxed hammer within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination and test shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3). The preproduction pack may be accomplished utilizing either the preproduction hammer or the production hammer. If the preproduction model hammer is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government when requested by the Government to facilitate comparison between the preproduction model hammer and the production hammer.

Preservation. Preservation shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Disassembly. Disassembly of the hammer shall be the minimum necessary to safeguard parts vulnerable to damage or loss and to accomplish reduction in cube, in accordance with the manufacturer's recommendations. Removed bolts, nuts, screws, washers, and pins shall be reinstalled in mating parts to prevent their loss. Removed keys shall be secured to the keyway of the primary component or shaft.

5.2.1.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.1.3 Unprotected metal surfaces. Unprotected exterior metal surfaces not specified herein requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative.

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5.2.1.8 Instrument panel. Instrument glasses not protected by a metal housing or cover shall be protected with a fitted piece of hardboard or plywood, applied over the individual glasses or the entire panel, and secured in place with tape conforming to MIL-T-22085, Type I or II.

5.2.1.9 Fire extinguisher. Unpainted exterior surfaces of the fire extinguisher requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with Type P-1 preservative. The fire extinguisher shall then be placed in and secured in the bracket furnished.

5.2.1.10 Repair parts. The preservation application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve the repair parts.

5.2.1.11 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, Method IC-1 or IC-3 and placed in the toolbox.

5.2.1.12 Maintenance tools. Maintenance tools shall be preserved in accordance with the Level A requirements of PPP-P-40 and placed in the toolbox.

5.2.1.13 Consolidation. Preserved and unpreserved small loose components that cannot be placed in the toolbox and that require the protection afforded by a box to prevent damage, pilferage, or loss shall be consolidated together in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional, Grade B, or PPP-B-621, Class 2, style optional. The boxes shall be of a size most convenient for securing to the hammer or to the base of the box specified in 5.3.1 without increasing cube. The contents shall be blocked, braced, and anchored or cushioned as applicable to prevent free movement or damage. Box closure shall be in accordance with the appendix to the applicable box specification. Strapping shall not be required. Large components, such as the turntable and setoff rails shall be secured to the hammer as provided for, or to the base specified in 5.3.1 with steel strapping conforming to QQ-S-781, class and type optional, Finish A, size as applicable, in a manner to prevent movement or damage.

5.2.2 Level B. Each complete hammer shall be preserved as specified in 5.2.1 for Level A, with the following exceptions.

5.2.2.1 Tags. Information tags shall conform to UU-T-81, type optional and the ink required for applying the information on the tags is not required to be waterproof.

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5.2.2.2 Drive chain. Exposed drive chains shall be coated only with Type P-3 or P-9 preservative.

5.2.2.3 Enclosed gears. Gear housings shall be filled to the operating level with the applicable type lubricant required for operation. Tags shall indicate: "The housing is filled with operational lubricant. Do not drain until the first recommended lubricant change."

5.2.2.4 Consolidation. Boxes for consolidation shall be Domestic Type or Class as applicable.

5.2.3 Commercial. Each complete hammer and all items furnished with the hammer shall be preserved to afford protection against mechanical and physical damage as follows: Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coatings, volatile corrosion inhibitors or desiccated packs. Items requiring protection from physical damage shall be protected by wrapping, or cartonizing of the individual item. Liquid cooling system shall be filled with a fresh clean solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153, and a tag attached indicating the temperature to which the cooling system can be subjected before damage. All lubrication reservoirs, sumps, etc., shall be filled to the operating level with the grade of military lubricant designated for use in the temperature range at the shipping destination. A tag shall be attached in a conspicuous location to indicate which lubricant has been used. A deprocessing guide shall be furnished indicating in detail the preservation material to be removed and any other servicing required prior to placing the hammer in operation.

5.3 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.3.1 Level A. Each complete hammer, preserved as specified in 5.2 shall be packed in a box conforming to MIL-B-26195, Type II, style optional, Class 2, with rubbing strips and plymouth superstructures. Blocking, bracing, and anchoring of the contents shall be in accordance with MIL-STD-1186.

5.3.2 Level B. Each complete spike hammer, preserved as specified in 5.2, shall be packed as specified in 5.3.1 except the box shall be Type I.

5.3.3 Level C. Each complete spike-hammer, preserved as specified in 5.2, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

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5.4 Preservation and depreservation guide. Two copies of the preservation and depreservation guide for railroad equipment, conforming to DA Form 3257 shall be prepared. One copy shall be placed in the package of technical publications and the package marked to indicate: "Preservation and Depreservation Guide enclosed". The other copy shall be placed in a sealed waterproof envelope and marked "Preservation and Depreservation Guide" then secured in a conspicuous location on the spike hammer.

5.5 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The hammer is intended for use in driving spikes into railroad ties.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and class of hammer required (see 1.2).
(c) Time frame required for submission of the preproduction pack (see 5.1).
(d) Degree of preservation and packing required (see 5.2 and 5.3).

6.3 Preproduction pack. Any changes or deviations of production packs om the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the hammer in accordance with this specification.

6.5 Government serial numbers. The contracting officer should furnish the Government serial numbers for the hammers upon award of the contract or order.

6.6 Lubricants. The contracting officer should furnish a list of lubricants applicable to the hammer covered by this specification with the commercial equivalent when requested by the contractor.

6.7 First-produced hammers. Any changes or deviations of production hammers from the approved first-produced hammers during production will be subject to the approval of the contracting officer. Approval of a first-produced hammer will not relieve the contractor of his obligation to furnish hammers conforming to this specification.

6.8 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operational manual to be furnished with each hammer.

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6.9 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary including any special tools and instructions on shipment of hammers.

6.10 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.3).

6.11 DA Form 3257. The contracting officer should arrange to furnish DA Form 3257, Preservation and Depreservation Guide for Railroad Equipment, when requested by the contractor.

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DOCUMENT IDENTIFIER AND TITLE

MIL-H-52265B(ME) Hammers, Track Spike, Gasoline- or Diesel-Engine-Driven

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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